

Annular Fuel Heat Transfer Derivation and DeepONet Surrogate Modeling Plan

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1 Annular Fuel Heat Transfer Derivation

1.1 Introduction

Here is a proposed derivation for heat transfer in an annular fuel segment, described in Figure 1. There is an inner coolant with nominal temperature T_i and heat transfer coefficient htc_i in the region $0 \leq r < r_i$, a fuel region with a constant volumetric heat generation rate (VHGR) q_0''' in the region $r_i \leq r \leq r_o$, and outer coolant region $r > r_o$ with nominal temperature T_o , heat transfer coefficient htc_o . Note the model can generalize to gas gap and cladding by taking $htc = htc_{conv} + htc_{clad} + htc_{gap}$ on the inner and outer surfaces.

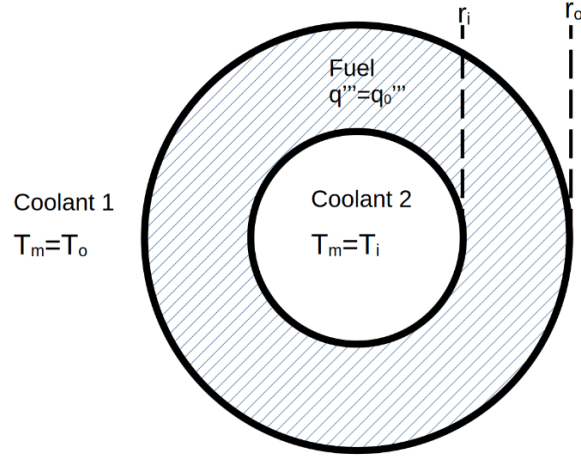


Figure 1: Domain of annular problem

The radial heat equation, allowing for a fuel conductivity $k_f = k_f(T)$ that depends on temperature, can be given by:

$$\frac{1}{r} \frac{d}{dr} \left[r k_f(T) \frac{dT}{dr} \right] = -q'''(z) \quad (1)$$

Rearranging (multiplying by r and integrating twice):

$$\int k_f(T) dT = -\frac{q'''(z)}{4} r^2 + C_1 \ln(r) + C_2$$

Let $\Theta[k_f](T) = \int_0^T k_f(T') dT'$ denote the conductivity integral functional, which is a function of the fuel temperature. We have:

$$\Theta[k_f](T) = -\frac{q'''(z)}{4} r^2 + C_1 \ln(r) + C_2 \quad (2)$$

The heat flux is given via Fourier's law:

$$-k_f(T) \frac{dT}{dr} = q''(r)$$

Thus,

$$q''(r) = \frac{q'''(z)}{2}r - \frac{C_1}{r}$$

The iteration scheme assumes an initial guess for the inner and outer heat fluxes in the annular region. With the heat flux guess, the inner and outer fuel temperatures can be computed by solving the governing heat transfer equations from the bulk coolant to the outer fuel temperatures. For heat transfer coefficient given generally by $htc = \frac{1}{htc_{conv}} + \frac{1}{htc_{clad}} + \frac{1}{htc_{gap}}$, the fuel temps at r_i, r_o can be obtained via the following equation, where the index j indicates either the inner or outer surface.

$$T_{fo}(r_j) = T_{m,j} + \frac{1}{htc_j} q''_j$$

The assumed guesses for heat fluxes lead to guesses for the fuel surface temperatures on the inner and outer surfaces. Then, we obtain 2 equations from the heat equation solution at r_i and r_o , evaluating the conductivity integral forward at the (known, guessed) surface temperatures $T_{fo}(r_i), T_{fo}(r_o)$:

$$\Theta[k_f](T_{fo}(r_i)) = -\frac{q'''(z)}{4}r_i^2 + C_1 \ln(r_i) + C_2$$

$$\Theta[k_f](T_{fo}(r_o)) = -\frac{q'''(z)}{4}r_o^2 + C_1 \ln(r_o) + C_2$$

Which is a 2x2 linear system for C_1 and C_2 , since $\Theta[k_f](T_{fo}(r_j))$ is a known number once the flux guess fixes $T_{fo}(r_j)$ — this is the key advantage of the scheme: it only ever needs Θ evaluated forward (as an integral of k_f), never inverted. Writing $\Theta_j \equiv \Theta[k_f](T_{fo}(r_j))$ and $\alpha_j \equiv \Theta_j + \frac{q'''(z)}{4}r_j^2$, the system is:

$$\ln(r_i) C_1 + C_2 = \alpha_i, \quad \ln(r_o) C_1 + C_2 = \alpha_o$$

This has an analytic solution using Cramer's rule:

$$C_1 = \frac{\alpha_i - \alpha_o}{\ln(r_i) - \ln(r_o)} = \frac{[\Theta_i - \Theta_o] + \frac{q'''(z)}{4}(r_i^2 - r_o^2)}{\ln(r_i/r_o)}$$

$$C_2 = \frac{\ln(r_i) \alpha_o - \ln(r_o) \alpha_i}{\ln(r_i) - \ln(r_o)}$$

With the new guess value for C_1 , new guess values for the heat fluxes can be obtained directly from the flux relation above (note only C_1 , not C_2 , is needed here):

$$q''_{i,new} = \frac{q'''(z)}{2}r_i - \frac{C_1}{r_i}, \quad q''_{o,new} = \frac{q'''(z)}{2}r_o - \frac{C_1}{r_o}$$

This iteration will continue until it converges to the true heat flux distribution.

1.2 Considerations for numerical methods

Figure 2 summarizes the iteration scheme derived in Section 1.1 for solving the coupled fuel temperature and heat flux distribution.

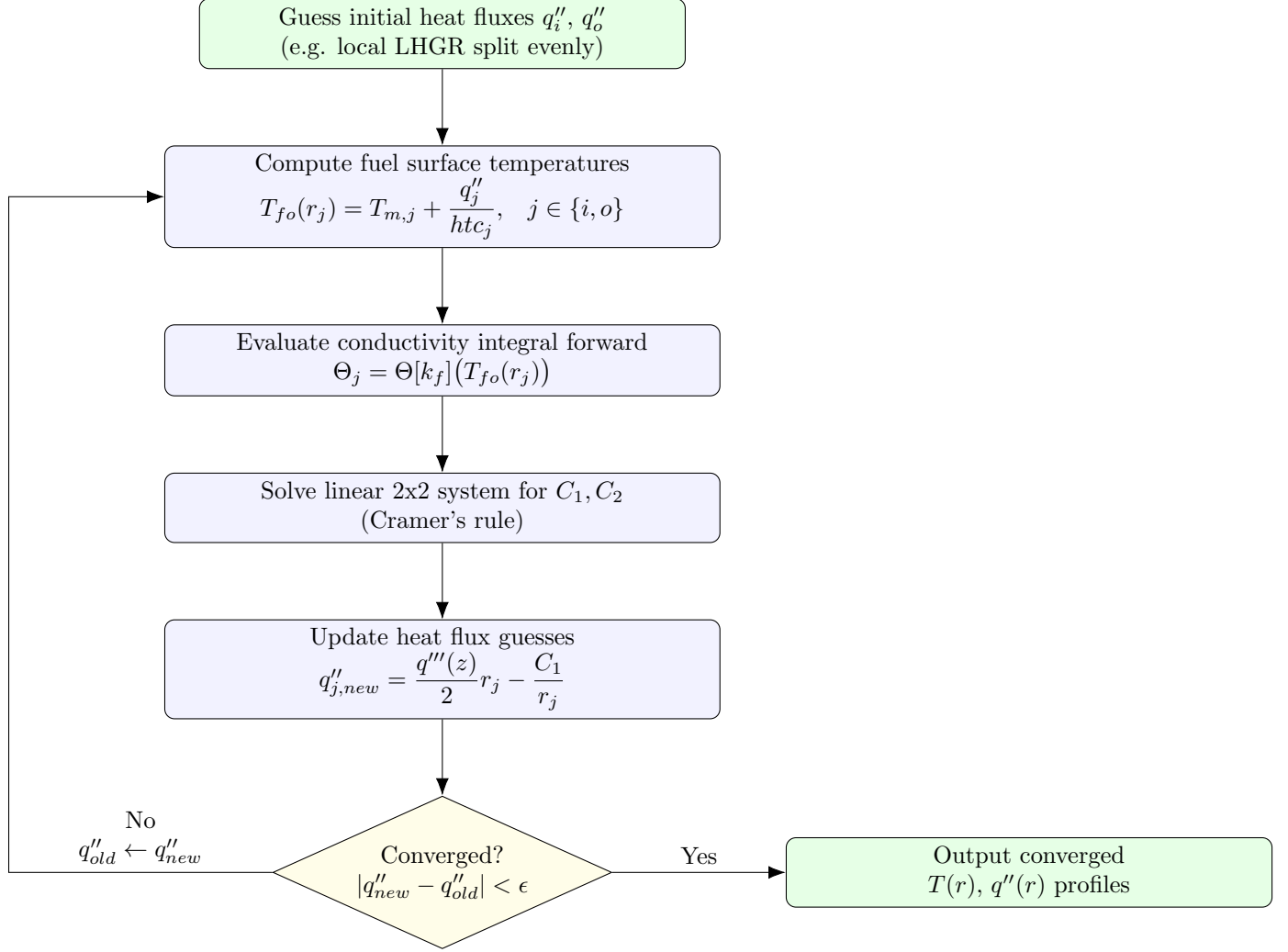


Figure 2: Flow diagram of the iterative scheme for solving the coupled fuel temperature and heat flux distribution.

In the case of supercritical water, most available correlations for heat transfer coefficient require knowledge on the wall temperature and are thus implicit. The computational scheme is explained in the figure. The iteration scheme will start with a guess value for the inner and outer heat fluxes; this guess value will be the local LHGR divided by 2. Potentially, a better guess value could be obtained using information from the last step. Let $f = \frac{q'_i}{q'_o}$ denote the heat ratio, Using the fraction from the last axial step, the new guess values could be:

$$q'_{o,guess}(z_j) = q'(z_j) \cdot \frac{1}{1 + f_{j-1}}$$

$$q'_{i,guess}(z_j) = q'(z_j) \cdot \frac{f_{j-1}}{1 + f_{j-1}}$$

With the heat guess values, the Swenson correlation can be used to determine htc and the surface temperature. For surface k:

$$q''_k = \frac{q'_k}{2\pi R_{co}} = htc_{Sw}(T_m, T_w) \cdot [T_w - T_s]$$

So solving iteratively for the wall temperature T_w

$$f_{wall}(T_w) = \frac{q'_k}{2\pi R_{co}} - htc_{Sw}(T_m, T_w) \cdot [T_w - T_s] = 0, T_w = root(f_{wall}, T_w)$$

Then, the following equation can be used to determine the cladding temperature:

$$T_{ci}(z_i) = T_{co}(z_i) + \frac{q'_k(z_i)}{2\pi k_c(z)} \cdot \ln \left(\frac{R_{co}}{R_{ci}} \right)$$

2 DeepONet/PINN Plan

2.1 Introduction

A deepoNet is a type of neural operator that maps some input space and a function, to another function. For example, a deepoNet could be used to learn the solution to a non homogenous differential equation for any general forcing function. For the SCA problem, we have multiple outputs for axial enthalpy, temperature, etc that depend on the geometric and input parameters.

The balance equation can be given by:

$$G \frac{dh}{dz} = \frac{1}{A_z} q'(z) + \frac{G}{\rho} \left[\frac{dP}{dz} + f \frac{G|G|}{2D_h \rho} \right] \quad (3)$$

Or neglecting the energy from pressure gradient and frictional losses:

$$G \frac{dh}{dz} = \frac{1}{A_z} q'(z) + \frac{G}{\rho} \quad (4)$$

Or rearranging:

$$\frac{dh}{dz} = \frac{q'(z)}{\dot{m}}$$

$$h \rightarrow h^* \cdot h_{in} z \rightarrow z^* \cdot L q'(z) \rightarrow F_q(z) \cdot \langle q' \rangle$$

Non-Dimensionalizing:

$$\begin{aligned} \frac{d(h^* h_o)}{d(z^* L)} &= \frac{F_q(z) \langle q' \rangle}{\dot{m}} \\ \frac{dh^*}{dz^*} &= \frac{\langle q' \rangle L}{\dot{m} h_0} F_q(z^* L) \end{aligned}$$

Let the power ratio, F_q , be some periodic function with a Nth order Fourier expansion squared (in order to ensure positivity), with an offset ϕ_q , to ensure the profile never reaches zero. Let $S(x)$ denote the bare Fourier sum:

$$S(x) = \sum_{n=1}^N \left[a_n \cos \left(\frac{\pi n x}{L} \right) + b_n \sin \left(\frac{\pi n x}{L} \right) \right]$$

so that

$$\begin{aligned} F_q(x) &= S(x)^2 + \phi_q \\ F_q(x^* L) &= S(x^* L)^2 + \phi_q, \quad S(x^* L) = \sum_{n=1}^N [a_n \cos(\pi n x^*) + b_n \sin(\pi n x^*)] \end{aligned}$$

The average heating profile can be given by:

$$\langle q' \rangle = \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} F_q(z) dz = \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} [S(z)^2 + \phi_q] dz = \frac{1}{L} \langle S, S \rangle + \phi_q$$

Since S is expanded in an orthogonal basis on $[-L/2, L/2]$ — with $\int_{-L/2}^{L/2} \cos^2(\pi n z/L) dz = \int_{-L/2}^{L/2} \sin^2(\pi n z/L) dz = L/2$ and all cross terms (different n , or cos against sin) vanishing — the inner product reduces to the sum of the squares, giving an explicit form:

$$\langle q' \rangle = \frac{1}{2} \sum_{n=1}^N [a_n^2 + b_n^2] + \phi_q$$

For the annular problem, we have a system of coupled ODEs, with a closure equation G that returns the inner and outer heat fluxes given the inner and outer enthalpy profiles,

$$\frac{dh_i^*}{dz^*} = \frac{\langle q'_i \rangle L}{\dot{m} h_0} F_{q,i}(z^* L)$$

$$\frac{dh_o^*}{dz^*} = \frac{\langle q'_o \rangle L}{\dot{m} h_0} F_{q,o}(z^* L)$$

$$q'_o, q'_i = G(h_o, h_i)$$

The closure can be given by the conduction and heat equation.

For the consideration of a PINN to solve this problem, the power profile must be specified, and then the loss function would include the heat balance equation and the implicit heat transfer equations. If it is ensured that proper constraints are made to ensure the positivity of the network outputs and other constraints, its possible that the network can resolve the implicit equations directly, avoiding the need for running solver routines explicitly for every SCA input case.

Additionally, is also possible that a regular PINN can be trained with expansion coefficients (either for fourier, Legendre, Chebyshev, etc) can be added as inputs to the PINN. This way, the PINN can natively generate an unlimited number of solutions for different power profiles, bypassing the need to generate a large number of datasets using direct numerical techniques.

2.2 Model Inputs

It is important to specify the exact inputs that will be given to the deeponet. One important thing to consider is the mass flux in the inner and outer regions of the pin. This is a parallel system, so by training the deeponet with separate inputs for G_o and G_i , and iterative solver and be used on the trained model to enforce equal pressure drop, and $G_{in} = G_o + G_i$ given some specified total inlet mass flux.

Some important input parameters are given in table 1:

Symbol	Definition
$r_{co,i}$ or r_i	Outer pin radius
$r_{co,o}$ or r_o	Inner annulus radius
t_{ci}	Inner cladding thickness
t_{co}	Outer cladding thickness
δ_i	Inner annulus region gas gap
δ_o	Outer annulus region gas gap
$Pitch$	Pitch
G_i, G_o	Mass flux in inner and outer region
L	Active Fuel Region (AFR) length
$\langle q' \rangle$	Average LHGR

Table 1: Definition of Variables and Parameters

It will be of importance to determine if an explicit pinn can be used without data from finite difference solvers, or if a combination of the data from a tradition solver and the physics based loss is nescessary.

2.3 Other Misc Things

The iterative possibilities are high with a working deeponet. Add more inputs for burnup, maybe flux or dpa, and add some conductivity models for the cladding and gas gap, othper parameters. Rapid iteration can

be used to determine heated gap distance, or cladding conductivity models. Furthermore, this necessitates adding correlations for thermal expansion, fuel swelling, and cladding conductivity. Make data gen routines that toss away bad or unconverged inputs, implement a decent user interface for specifying models and such.